BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 4461 / 4462

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

Q. 1223 (Inter Part - II) — Biology (Objective) — Session 2019-21 to 2021-23 — Group I — Paper (II)

Q. Time Allowed:- 20 minutes — Maximum Marks:- 17 — PAPER CODE 4461

Note:- You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of Ink Remover or white correcting fluid is not allowed.

Q. Q. 1

Q1. Which is not a Poikilotherm?

(A) (B)

(C) (D)

Q2. Fibrous Tissue which connects two bones is called

(A) (B)

(C) (D)

Q3. Posterior four vertebrae of pelvic region form

(A) (B)

(C) (D)

Q4. Which type of Neuron have long axon?

(A) (B)

(C) (D)

Q5. All are phases of menstrual cycle except

(A) (B)

(C) (D)

Q6. In the zone of elongation, the volume of the cells increases up to =

(A) (B)

(C) (D)

Q7. Which neurotransmitter secreted at synapse outside the central nervous system:

(A) (B)

(C) (D)

Q8. Which of the followings is required for joining okazaki fragments during DNA Replication?

(A) (B)

(C) (D)

Q9. Contraction of spindles occur during:

(A) (B)

(C) (D)

Q10. Anticodons are present on =

(A) (B)

(C) (D)

Q11. Interaction between genes occupying different Loci is:

(A) (B)

(C) (D)

Q12. pSC 101 has antibiotic resistant gene for

(A) (B)

(C) (D)

Q13. Patients of cystic fibrosis die due to numerous infections of the

(A) (B)

(C) (D)

Q14. Endosymbiont Hypothesis explains origin of

(A) (B)

(C) (D)

Q15. All the populations within an ecosystem are known as

(A) (B)

(C) (D)

Q16. Coniferous forest located at high latitude are called

(A) (B)

(C) (D)

Q17. The decline in thickness of ozone layer is caused by increasing level of

(A) (B)

(C) (D)

> Note: Board typos kept verbatim: 'okazaki', 'Nitrocarbons'.

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

Q. 1223 (Inter Part-II) — Biology (Subjective) — Session 2019-21 to 2021-23 — (Group I) — Paper (II)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) What are the adaptations of xerophytes for Osmoregulation? (ii) Differentiate b/w Osmoconformers and osmoregulators? (iii) What is the role of ADH and aldosteron in Osmoregulation? (iv) Differentiate b/w collenchyma and sclerenchyma. (v) Write the name of Bones of Cranium. (vi) What is the effect of exercise on Muscle? (vii) How is a seed formed? (viii) What is the role of corticosteroid in birth process? (ix) What are the adaptations in plants and animals for terrestrial ecosystem? (x) Compare the rain fall in Temperate decidous forests and Grassland ecosystem. (xi) What is the role of soil for plants and animals? (xii) How we can save energy?

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Differentiate homozygous and heterozygous conditions. (ii) Why is Pituitary anterior lobe referred to as Master gland? (iii) Define Habituation in terms of animal behaviour. (iv) What is sympathetic Nerve system? (v) Define Pleiotropy. Give one example. (vi) If recombination frequency is 20% then draw a gene map. (linkage map). (vii) How is genomic library made?

(viii) Give an application of Transgenic bacteria. (ix) How did scientists produce a salt-tolerant plant? (x) Define Autecology. (xi) How do root nodular bacteria give and take benefits during symbiotic association? (xii) Differentiate Macronutrients and Micronutrients.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) What are teratogens? Give examples. (ii) Lateral buds in plants can be released from the effect of Apical bud. Comment on it. (iii) Discuss the bondings which hold together two strands of DNA in double helix. (iv) How two strands of DNA get synthesized during DNA replication? (v) How RNA polymerase form Transcription bubble on a gene? Discuss it. (vi) What is Metastasis? (vii) Distinguish Apoptosis from Necrosis. (viii) What are vestigical organs? Give examples. (ix) Define Endosymbiont hypothesis.

Q. (Marks 8 × 3 = 24) Section ---------- II Note: Attempt any three questions.

Q5. 5. (a) Draw Labelled diagram of vertebrate Nephron. State the function of each part. (b) Define mitosis, only explain its importance.

Q6. 6. (a) Highlight the types of directional responses in plants which are caused due to external stimuli. (b) Define Xerosere. Describe the stages of Xerosere

Q7. 7. (a) Explain how reflex action prevent body damage during emergency? (b) Define endangered species. Discuss causes of extinction and conservation plan.

Q8. 8. (a) What is Diabetes mellitus? Explain its genetic basis. (b) Describe the human male reproductive system.

Q9. 9. (a) Explain in detail the phenomenon of Growth Correlation with example. Write its commercial application. (b) Explain the process of polymerase chain reaction with the help of diagram.

> Note: Board typos kept verbatim: 'aldosteron', 'Temperate decidous', 'Autecology', 'bondings', 'vestigical'.

GROUP – II

Q. Paper : II (Objective Type)

سوالیہ پرچہ: II (معروضی طرز)

Q. 1223 (Inter Part - II) — Biology (Objective) — Session 2019-21 to 2021-23 — (Group 2nd) — Paper (II)

Q. Time Allowed:- 20 minutes — Maximum Marks:- 17 — PAPER CODE 4462

Note:- You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of Ink Remover or white correcting fluid is not allowed.

Q. Q. 1

Q1. Fever causing substance is called.

(A) (B)

(C) (D)

Q2. Brain is protected by

(A) (B)

(C) (D)

Q3. Action of venus fly trap is:

(A) (B)

(C) (D)

Q4. Hormone that supresses ovulation is

(A) (B)

(C) (D)

Q5. Fruit ripening is due to production of

(A) (B)

(C) (D)

Q6. Diploid parthenogenesis may occur in

(A) (B)

(C) (D)

Q7. Acetabularia is unicellular:

(A) (B)

(C) (D)

Q8. Initiation codes for every protein coding gene is AUG which encodes for

(A) (B)

(C) (D)

Q9. Histone proteins combine with DNA to form a complex.

(A) (B)

(C) (D)

Q10. The S-phase of cell cycle takes

(A) (B)

(C) (D)

Q11. ABO blood group was discovered in:

(A) (B)

(C) (D)

Q12. pSC 101 plasmid has antibiotic restriction gene for

(A) (B)

(C) (D)

Q13. PCR technique was developed by

(A) (B)

(C) (D)

Q14. Who Published an essay on "The principle of population"?

(A) (B)

(C) (D)

Q15. Lithosphere includes

(A) (B)

(C) (D)

Q16. In grassland ecosystem tropical climates have woody trees called.

(A) (B)

(C) (D)

Q17. Which of the following act as environmental buffer?

(A) (B)

(C) (D)

> Note: Board typos kept verbatim: 'supresses', 'Progestrrone', 'Ampiciline', 'Streptomycine', 'Penecillin'.

Paper : II (Essay Type)

پرچہ: II (انشائیہ طرز)

Q. 1223 (Inter Part-II) — Biology (Subjective) — Session 2019-21 to 2021-23 — (Group 2nd) — Paper (II)

Q. Time Allowed: 2.40 hours — Maximum Marks: 68

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Write the metabolic waste products of amino acids and nucleic acids. (iii) Give the role of Juxtamedullary nephrons. (ii) What are heterotherms? Give two examples. (iv) Differentiate between Heartwood and Sapwood (v) What is nutation. (vi) What is ligament? Give its function. (vii) Define Climactric. (viii) Give the effect of red light on internode growth and etiolation. (ix) Differentiate between alpine and boreal forests. (x) What do you know about Pampas and Prairies? (xi) Give the distribution percentage of water in different forms on the earth. (xii) Define bioconversion

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) What are biological Rhythms? (ii) Write two commercial applications of Abscisic Acid. (iii) How is reflex action helpful in our daily life? (iv) Compare Gross and Net Primary Production. (v) What are compound Sex chromosomes? Write an example. (vi) What is product rule? (vii) Why was test cross devised by Mendel? (viii) Write the use of cell suspension culture technique. (ix) How do genetic engineers were able to produce salt tolerant plant "Arabidopsis". (x) What is the role of suicide gene in transgenic bacteria? (xi) What is Predation? Give its importance. (xii) Write the significance of root nodules in plants, Give an example.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) What role is played by gray-crescent in development? (ii) How gerentology is helping in increasing expected ages? (iii) Why DNA replication is semi-conservative? (iv) What do you know about benign and malignant tumors? (v) Why some portion of chromatin is condensed permanently and other remain condensed only during cell division? (vi) What is genetic code and also give its properties? (vii) Differentiate between apoptosis and necrosis. (viii) What is the theory of special creation? (ix) How biogeography does provide evidence for evolution?

Q. (Marks 8 × 3 = 24) Section ---------- II Note: Attempt any three questions.

Q5. 5. (a) Describe the process of concentration of excretory products in Human Nephron. (b) In what respect can cell death be regarded beneficial.

Q6. 6. (a) Describe a Hinge Joint and explain how it is moved by antagonistic muscles? (b) Describe flow of Energy in food chain of an ecosystem.

Q7. 7. (a) Explain types and functions of hormones released from cortex of Adrenal gland. (b) Give detailed evidences of evolution from comparative anatomy.

Q8. 8. (a) Describe vernalization in plants by giving its significance. (b) Describe the inheritance of a trait in which the phenotype of a heterozygote is intermediate between phenotypes of the two homozygotes by giving one example.

Q9. 9. (a) Describe the impact of Nucleus on development by giving example of Acetabularia. (b) How Severe Combined Immunodeficiency syndrome (SCID) children are treated

> Note: Board typos kept verbatim: 'Climactric', 'gerentology'.